

Abundance estimation and capture of sablefish (*Anoplopoma fimbria*) by longline gear

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Abstract: Longline surveys in Alaska measure sablefish (*Anoplopoma fimbria*) relative abundance and are the primary information source used for abundance and quota estimation. Hook timer, on-bottom (soak) time, hook density, hook pattern, bait type, and bait condition experiments and mathematical models were used to evaluate the performance of the longline surveys for estimating sablefish relative abundance. The rate that sablefish encountered the longline gear decreased with on-bottom time independently of sablefish density in the sampled area. Sablefish were adept at locating available baits, even when few remained. The decrease in encounter rate appears related to odor concentration at the leading edge of the odor plume. The ability to locate baits, even when few remain, differs from previous models of fish capture by longline in which the probability that a fish located a bait was proportional to the number of available baits. Decreased encounter rate and the ability to locate baits efficiently imply that longline catch rates likely provide an accurate index of fish abundance if the on-bottom time is long enough to cover the period when most fish encounter the gear and the initial bait density is high enough that baits remain available throughout the soak; the weak link between catch rate and abundance is the unknown extent that factors such as temperature and food availability affect the proportion of fish caught.

Résumé : Les relevés à la palangre menés en Alaska pour mesurer l'abondance relative de la morue charbonnière, *Anoplopoma fimbria*, constituent la première source d'information pour l'estimation de l'abondance et l'établissement des quotas. Nous nous sommes servis d'expériences sur le temps nécessaire à la capture, la durée de séjour sur le fond (mouillage), la densité d'hameçons, le mode d'accrochage des hameçons, le type d'appât et l'état des appâts, ainsi que de modèles mathématiques, pour évaluer l'intérêt des relevés à la palangre pour l'estimation de l'abondance relative de la morue charbonnière. Le taux de rencontre de la morue charbonnière et de la palangre baissait en fonction de la durée de mouillage, indépendamment de la densité de ce poisson dans la zone d'étude. Les morues charbonnières étaient promptes à localiser les appâts disponibles, même quand ils étaient peu nombreux sur la ligne. La baisse du taux de rencontre semble liée à la concentration de l'odeur au bord avant du panache d'odeur. La capacité de localiser les appâts, même quand il en reste peu sur la ligne, introduit un nouvel élément par rapport aux modèles antérieurs de capture des poissons à la palangre, dans lesquels la probabilité qu'un poisson trouve un appât était proportionnelle au nombre d'appâts disponibles. La baisse du taux de rencontre et la capacité de localiser efficacement les appâts indiquent que les taux de capture à la palangre peuvent vraisemblablement fournir un indice précis de l'abondance des poissons si la durée de mouillage sur le fond est assez longue pour couvrir la période où la plupart des poissons rencontrent l'engin, et si la densité initiale d'appâts est assez élevée pour qu'il reste des appâts tout au long de la durée de mouillage; il reste un maillon faible entre le taux de capture et l'abondance, du fait qu'on ne sait pas dans quelle mesure des facteurs comme la température et la disponibilité de la nourriture affectent la proportion de poissons capturés.

[Traduit par la Rédaction]

Introduction

Abundance indices generally are determined from catch and effort data of a fishery or research surveys. Understanding the catch rate – abundance relationship and, more generally, the interaction of the target species with the gear is essential for resource assessment (Gunderson 1993). Although nonlinear relationships have been proposed (Bannerot and Austin 1983; Richards and Schnute 1986), the most common hypothesis has catch rate, U , proportional to

abundance, $U = qN$, where q is the catchability coefficient and N is abundance. If true, a 10% increase in U is expected to represent a 10% change in abundance. Sablefish (*Anoplopoma fimbria*) relative abundance has been estimated by annual longline surveys in Alaska since 1978 (Sasaki 1985; Sigler and Fujioka 1988; Sigler and Zenger 1989). A better understanding of sablefish capture by longline gear was necessary to determine whether these surveys accurately index sablefish abundance.

Two parameters affecting fish capture by longline gear are the rate that fish encounter the gear (number of fish per unit on-bottom time) and, after encountering the gear, the probability that fish successfully locate an available bait. I conducted a series of field experiments to measure sablefish encounter rate and their ability to locate bait. Capture times were observed during hook timer experiments and used to estimate encounter rate and bait location probability. Soak time experiments were conducted to examine how catch rate,

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a measure of encounter rate, changed with on-bottom time. Hook density and hook pattern experiments were conducted to examine how bait density affected catch rate, a measure of the ability to locate bait. Bait type and bait condition experiments were conducted to examine how odor concentration affected catch rate, a measure of the relationship of odor concentration and encounter rate.

In this paper, I develop a mathematical model to analyze the hook timer experiments. The model also provides a conceptual framework to discuss the results of all the experiments. The model describes three events in fish capture: encountering gear, locating a bait, and hooking. In the model, encounter rate may vary as a function of time, bait location probability may vary as a function of the number of available baits, and hooking fraction is assumed constant. Mathematical models previously developed to describe gear saturation (Murphy 1960; Rothschild 1967) are a special case of my model and also were tested. In fisheries literature, "gear saturation" (Beverton and Holt 1957) describes bait location probability as "the effect of fish density to reduce the fishing power of a unit of fishing gear as the catch in it increases."

Methods

Hook timer experiments

Capture times were collected to determine how gear saturation and encounter rate affect sablefish capture rates. Capture times were collected in two hook timer experiments conducted from 26 June to 2 July 1992 and from 24 May to 3 June 1993. A hook timer is an electromechanical device that records the capture time of a fish (Somerton and Kikkiwa 1995). An area of historically high sablefish catch rates, Chatham Strait, southeastern Alaska, was selected for the hook timer experiments because the motivation for these experiments was estimation of the effect of gear saturation, which is more likely to occur in an area of high catch rates like Chatham Strait. Two sites were sampled, one near Tenakee Inlet and one near Point Couverden.

A standard skate consisted of 100 m of line with 45 hooks spaced 2 m apart and baited with *Illex* squid, similar to the annual longline surveys. Hook timers were attached to every hook on the standard skates. Skates with 90 hooks spaced 1 m apart also were tested to determine the effect of hook density on capture times. On these skates, timers were attached to every other hook to reduce the number of timers that could be lost in a single set. A set at the Tenakee Inlet site consisted of two standard skates, with and without timers. A set at the Point Couverden site consisted of two skates, standard and 1-m spacing. Timers were attached to only one skate in a set to reduce the number of timers that could be lost in a single set. The hook density of the skate with timers alternated between sets.

A hook timer clock starts running when the timer attached to a baited hook is tripped. At retrieval, the following information was recorded electronically (Sigler 1994): timer status (tripped or untripped), hook status (fish species, baited, or bare), time elapsed since tripping, and time of day. Capture times were computed from this information and the time that the gear reached bottom, which was recorded with a time-depth recorder.

Some captures are not recorded by the hook timers. Observed captures signal their capture by tripping the timer. Observed captures include timers tripped on bottom and retrieved with hooks occupied by fish or bare. Considering bare hooks as captures presumes that a fish or invertebrate stole the bait and that, effectively, the hook is occupied because no bait remains to lure fish. Unob-

served captures do not trip the timer when they steal the bait or are caught. The capture time and order are known for the observed, but not the unobserved, captures. Unobserved captures include untripped timers, timers tripped at the rail, and defective timers retrieved occupied with fish or bare. These bare hooks are considered unobserved captures because the bait was removed, yet the timer was not tripped while on bottom to record the removal.

All hooks retrieved with bait were classified as baited hooks, regardless of timer status, because the bait remained available, even if attacked. Ineffective hooks (broken, bent, or tangled) were omitted from the calculations. Skates with more than five ineffective hooks for the 2-m spacing or 10 ineffective hooks for the 1-m spacing also were omitted.

The number of baits available when observed captures occurred is unknown because the capture time and order of the unobserved captures are unknown. The effect of the unobserved captures is approximated by assuming that the distributions of capture times and capture order are the same for observed and unobserved captures. Given N_b , the initial number of baits, N_o , the number of observed captures, and N_u , the number of unobserved captures, an estimate of the number of available baits when the i th observed capture occurred, x_i , is

$$x_i = N_b - (i - 1) \frac{N_o + N_u}{N_o}$$

which inflates the number of observed captures before the i th capture ($i - 1$) according to the ratio of total to observed captures.

Other longline gear experiments

Nine experiments on hook density, hook pattern, soak time, and bait type and condition also were conducted. All experiments including the hook timer experiments were conducted at depths of at least 250 m. In the hook density experiment, three densities of 1-, 2-, and 4-m spacing between hooks were tested. A set consisted of nine skates, three of each hook density. Bait was Pacific herring (*Clupea harengus pallasii*) instead of squid used in annual surveys. The hook density experiment was conducted from 4 to 13 July 1986 in Chatham Strait near Tenakee Inlet.

Hook density was found to affect sablefish catch rates, so two later experiments varied hook pattern while holding hook density constant. Distance between baits was tested by leaving various-sized gaps between sections with baited hooks; hook density was approximately 0.25 hook per metre of line for all patterns. One example of a hook pattern is three hooks spaced 2 m apart, 8 m of bare line, three hooks spaced 2 m apart, etc., denoted 8-m gap. Sections of line without hooks are equivalent to sections without baits, so these experiments also test for gear saturation. Two hook pattern experiments were conducted, one from 27 April to 5 May 1990 in Clarence Strait, Alaska, and the other from 22 to 26 July 1998 in the Gulf of Alaska near Yakutat Bay. In the first experiment, three hook patterns were tested, 4-m gap, 8-m gap, and 24-m gap. A set included six experimental skates. The experimental skates were located in two blocks, with one skate of each pattern in a block. The patterns were blocked to account for variation due to location, time, etc. In the second experiment, four hook patterns were tested, 12-m gap, 22-m gap, 32-m gap, and 42-m gap. A set included 64 experimental skates located in eight blocks with two skates of each hook pattern in a block. Wider gaps were tested in the second experiment after the first experiment failed to detect differences in catch rates among the first experiment's gap sizes. The order of pattern within a block was varied systematically in both experiments.

Four soak-time experiments were conducted in areas differing in sablefish abundance to determine whether catch rate increased with soak time and, if so, whether the relationship varied with abundance. The soak time experiments were conducted from 20 to 26

May 1986 in Chatham Strait near Tenakee Inlet, from 28 June to 3 July 1986 in Icy Strait, Alaska, near Home Shore, from 11 May to 1 June 1987 in Stikine Strait, Alaska, and from 5 to 26 March 1988 in Chatham Strait near Funter Bay. Two soak times, 3 and 7 h, were tested to approximate the shortest and longer soak times of the longline surveys (soak times commonly range to 10 h). Soak time is defined as the time between last buoy deployment and first buoy retrieval. A set consisted of 12 skates. Bait was herring instead of squid used in annual surveys, except for the last experiment, which tested both herring and squid. Several sites within an area were sampled during each experiment. When only herring was tested, each site was sampled twice, once per soak time, with the order of soak times varied systematically among sites to control for the effect of sampling order on catch rate. When both herring and squid were tested, four combinations resulted: herring, 3 h; herring, 7 h; squid, 3 h; squid, 7 h. Eight sites were sampled within the experimental area using a randomized block design. Each combination was tested at each site to account for the effect of site on the resultant catch rates. The order that the four combinations was tested at a site was varied systematically to control for the effect of sampling order on catch rate.

Two experiments with presoaked bait tested the importance of bait odor for attracting sablefish to the gear. The two experiments were conducted simultaneously during 9 August – 1 September 1991 in the Central Gulf of Alaska off Kodiak Island. In the first, baits were presoaked by placing tubs of baited and coiled longline gear into a tank supplied with fresh, flowing seawater. The hooks were baited before presoaking because baiting with squid is difficult when the bait is thawed rather than semifrozen. The duration of presoaking averaged 12 h. In the second, rather than presoaking artificially, reused baits were tested that had been soaked 3–10 h during a previous set. In both experiments, a set consisted of four experimental skates (either presoaked or reused bait) and 80 control skates with bait not soaked previously. Catch rates of the four experimental skates were compared with those of the adjacent four control skates.

Mathematical model

A mathematical model was fit to the capture time data. The model describes three events in fish capture: encounter the gear, locate an available bait, and become hooked. The encounter rate depends on odor plume drift and fish response. The downstream propagation of the odor plume depends essentially on advection; thus, the length of the odor plume at any time is computed as the product of current and time (Sainte-Marie and Hargrave 1987). Odor concentration at the plume's leading edge decreases quickly (Fig. 1). Løkkeborg et al. (1995) measured sablefish response intensity to different odor concentrations in a laboratory and found that response intensity decreased more slowly than odor dispersal (Fig. 1) and was most sensitive to feeding regimen (food ration and duration of food deprivation), more so than to current or fish speed.

Two encounter rate functions were tested, a power function and the exponential function. For the power encounter rate function, the encounter rate (λ) is assumed to be a function of the odor concentration at the leading edge of the odor plume with the response intensity described by Løkkeborg et al. (1995):

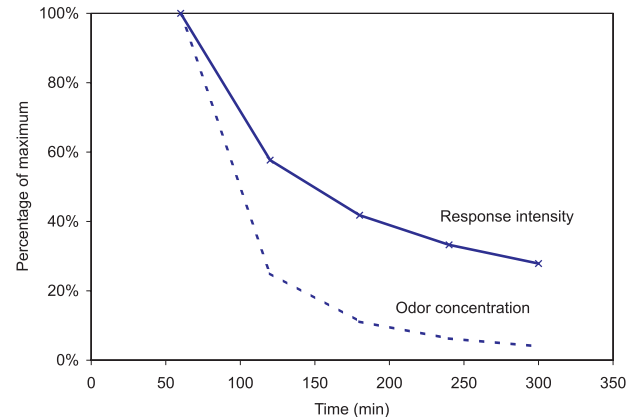
$$\lambda_t = \beta_3 10^{\beta_1 \log_{10} \frac{1}{t^2}}$$

See the Appendix for the derivation of this equation. The parameter β_1 is a measure of feeding regimen, the parameter β_3 is proportional to initial encounter rate, and t represents capture time.

For the exponential encounter rate function, the encounter rate (λ) is assumed to decrease exponentially with time at rate r :

$$\lambda_t = \lambda_0 e^{-rt}$$

Fig. 1. Odor concentration (broken line) (Sainte-Marie and Hargrave 1987; Løkkeborg et al. 1995) and sablefish response intensity ($\times$) (Løkkeborg et al. 1995) at the leading edge of the odor plume scaled to the maximum values versus time. The calculation equations for odor concentration and response intensity are shown in the Appendix. Response intensity is assumed to be proportional to the leading edge concentration. The assumed current was $0.08 \text{ m}\cdot\text{s}^{-1}$. The assumed sablefish feeding regimen was 1 day of food deprivation.



The parameter λ_0 is the initial encounter rate. The exponential function was tested because it well represents encounter rate data reported for whiting (*Gadus merlangus*) (Fernö et al. 1986, their fig. 3). The exponential encounter rate function is similar to the power encounter rate function, except that the power encounter rate function initially decreases more quickly.

After locating the gear, the fish may find an available bait. The bait location probability, p , is assumed to vary with the fraction of baits remaining, x . Initially, p is constant until the fraction of baits remaining has decreased to s , and thereafter, p decreases linearly to zero, i.e., $p_x = 1$ if $x \geq s$, $p_x = x/s$ if $x < s$. The value s is a measure of the fish's willingness to search for an available bait when one is not available at the point where the fish encounters the line. The value of s is estimated.

After locating an available bait, the fish may be hooked. The hooking fraction, h , is not measurable using capture times. I assumed that the hooking fraction was constant, implying that the behavioral or physiological states affecting the decision to attack an available bait and the rate of attack and escape are constant.

The capture rate, θ_{tx} , is the product of encounter rate, bait location probability, and hooking fraction: $\theta_{tx} = \lambda_t p_x h$. Thus, the fitted forms of the model are

$$\theta_{tx} = \beta_3 10^{\beta_1 \log_{10} \frac{1}{t^2}} p_x \text{ (power form)}$$

$$\theta_{tx} = \lambda_0 e^{-rt} p_x \text{ (exponential form)}$$

where $p_x = 1$ if $x \geq s$, $p_x = x/s$ if $x < s$. An encounter rate was estimated for each longline set, reflecting differences in local abundance. Bait location probability was estimated as one parameter for all sets assuming that bait location probability does not vary between sets.

A special case of the model was tested also:

$$\theta_{tx} = \lambda p_x \text{ (historic form)}$$

where $p_x = x$. This special case includes the longline models of Murphy (1960) and Rothschild (1967) for which encounter rate is constant and bait location probability is proportional to the fraction of baits remaining. For example, Murphy (1960) expressed the rate

of change of the number of hooked fish as a linear function of the number of available baits. For the historic form of the model, an encounter rate was estimated for each longline set, reflecting differences in local abundance. The bait location probability is fixed in the historic form but estimated in the exponential and power forms, i.e., the value of s is estimated in the exponential and power forms.

Expected total catch was computed by numerically integrating capture rate over time in 1-min intervals

$$\sum_t \theta_{\lambda_t}$$

where the fraction of baits remaining, x_t , evolves with time t . The parameters describing encounter rate and bait location probability were estimated by minimizing the negative log-likelihood, assuming that differences between the observed and corresponding expected catches were distributed normally. The log parameters rather than parameters on the original scale were estimated to improve reliability in the estimation process. This places all the parameters on a common scale and probably reduces parameter effects nonlinearity (Ratkowsky 1983).

The encounter rate estimates provide a measure of how much encounter rate affects survey results. Let b represent on-bottom time and b^* represent the maximum on-bottom time. A formula for the expected total number of encounters at b is found by integrating the exponential encounter rate function over on-bottom time and is

$$\frac{\lambda_0}{r} (1 - e^{-rb})$$

The proportion of encounters predicted to occur by b is

$$\frac{1 - e^{-rb}}{1 - e^{-rb^*}}$$

Results

Hook timer experiments

The timers were set successfully 19 times. Of the 845 successfully deployed timers, 136 tripped before or during setting of the longline gear or were defective, 472 tripped while the longline gear was on bottom, and 237 were retrieved untripped. Four hundred and sixty-five fish were caught and 180 timers were retrieved with bait. Most fish caught were sablefish (93%), which generally tripped the timer (85%). Sablefish size did not affect the chance of tripping the timer except for fish shorter than 50 cm fork length; however, these small fish composed only 8% of the sablefish catch. Two thirds of timers that tripped on bottom held a fish, demonstrating that more attacks strong enough to trip the timer result in capture than in escape. The escapes demonstrate that fish can attack a bait without becoming hooked, i.e., hooking fraction less than 1.0. Most (76%) untripped timers held no fish. For empty timers, hook status (baited, bare, or ineffective) was similar for timers tripped on bottom and untripped (contingency table analysis, χ^2 test, $p = 0.47$). The timers did not affect catch rate (19.3 ± 1.14 (SE) fish per skate without timers and 20.9 ± 1.38 with timers, paired t test, $p = 0.25$).

Capture rate decreases, slowly during early soak time, and substantially later (Fig. 2). Capture rate slowed substantially when 20–30% of the baits remained and after 20–100 min on bottom. The substantial slowing sometimes occurred smoothly (e.g., set 10) and sometimes abruptly (e.g., set 6).

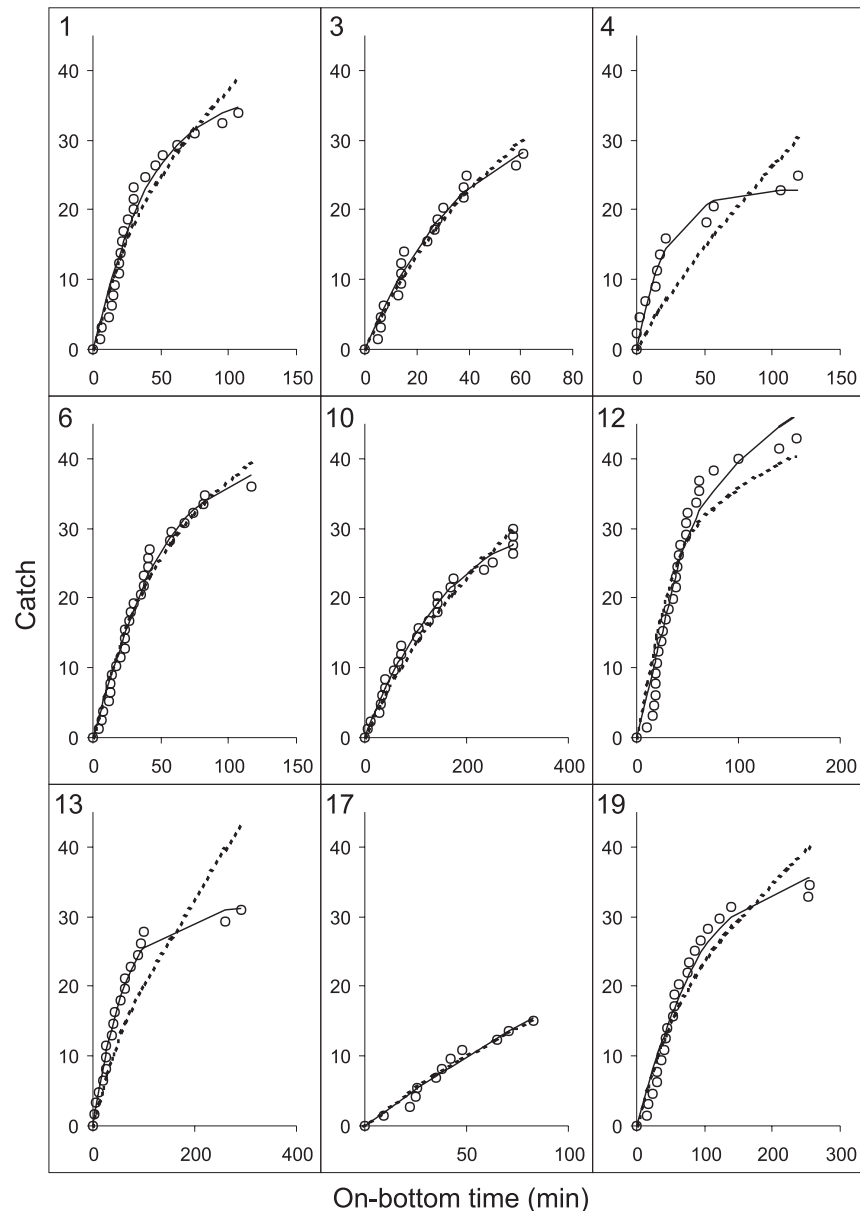
The historic longline model (constant encounter rate and bait location probability proportional to the number of baits remaining) represents the capture times fairly well (Fig. 2). Relaxing assumptions of the historic longline model by estimating time-varying encounter rate and by estimating bait location probability improved the fit substantially (Table 1) and represented the capture times better (Fig. 2). The better representation also markedly reduced the range of the residuals (Fig. 3). The representation is similar for the power and exponential models; only estimates for the exponential model are shown. The large negative residuals present early during on-bottom time for all three models (Fig. 3) are due mainly to data of two sets (e.g., set 12, Fig. 2) and imply that the observed encounter rate did not decrease so quickly as that estimated for these two sets.

For both the power and exponential models, the estimated bait location probability was constant until about half the available baits remained, contrary to the bait location probability assumed for the historic model (Fig. 4). The estimated encounter rate decreased with time for the power and exponential models (Fig. 5). The estimated encounter rates for these two models generally were similar for the first 75 min. Later, the encounter rate for the power function decreased slower than that for the exponential function. The exponential model better represented the capture time data based on a test comparing the alternative models (Table 1), the Akaike information criterion (Hilborn and Mangel 1997), so results are presented for the exponential model only. The estimated initial encounter rate, λ_0 , ranged from 0.07 to 1.23 encounters $\cdot \text{min}^{-1}$; the estimated instantaneous rate of decrease of the encounter rate, r , ranged from 0.000 to 0.047 $\cdot \text{min}^{-1}$ and averaged 0.011 $\cdot \text{min}^{-1}$ (Table 1). The average implies that after 1, 3, and 7 h, the encounter rate was 51, 13, and 1% of the initial rate. This also implies that at 1 and 3 h on bottom, 49 and 87% of fish were caught compared with the expected number caught by 10 h on bottom.

The log-likelihood surface for fitting the capture times does not curve smoothly to the global minimum for the exponential and power models; local minima were present in the surface where the minimization routine sometimes was trapped. Different starting values were used to locate and verify the global minimum. Choice of starting values was based on the observed capture times. For example, if initial captures occurred every 2 min, decreasing to every 10 min after 1 h, then reasonable starting values for λ_0 and r were 0.5 and $-\ln[(1/2)/(1/10)]/60 = 0.027$. The starting value for r was computed by solving the exponential encounter rate function for r , where 1/2 and 1/10 are the capture rates based on the observed capture times and 60 min is the time interval. The observed capture times also were a good check of the reasonableness of the resultant parameter estimates before declaring that a global minimum was found.

The capture rate can be estimated reliably, but because it is the product of λ_t and p_x , reliable estimation of the dependencies of λ_t and p_x on time and available baits is problematic. Both remaining baits and on-bottom time covary: the number of available baits decreases as fish take baits, which occurs through time as the gear is on bottom. For some sets, capture rate did not change as an obvious function of either time or available baits, limiting what can be inferred from these sets. This uninformative behavior happened for some

Fig. 2. Observed (○), expected exponential model (solid line), and expected historic model (broken line) catch for a 45-hook skate versus soak time. The models are described in the Mathematical model section. A constant capture rate is seen as a straight line. Slowing in capture rate is seen as a rightward drift. Off-bottom times are displayed in Fig. 6. The hook spacing of sets 1–13 was 2 m and of sets 14–19 was 1 m. Only a subset of the hook timer data is displayed for brevity.



sets where the gear was on bottom for only 1–2 h. Four sets were on bottom about 80 min each, and the capture rate was nearly constant during this interval for three of these sets (e.g., set 17, Fig. 2). The capture rate also was nearly constant at two sets with longer on-bottom times and then fell to zero for the remainder of the on-bottom time. These five sets (5, 15–18) had encounter rate and bait location probability that were nearly constant over the observed interval of available baits and time, so they provided little or no information about when or why the capture rate eventually slows because no substantial slowing was observed.

The general pattern of captures is similar regardless of hook density; initially, capture rate was nearly constant and then slowed substantially for both hook densities (e.g., com-

pare sets 1 and 19, Fig. 2). The initial encounter rate (0.61 for sets 14–19 versus 0.47 for sets 10–13, *t* test, one-tailed, $p = 0.23$, Point Couverden site) (Table 1) and the encounter rate's decline (0.0082 for sets 14–19 versus 0.0078 for sets 10–13, *t* test, one-tailed, $p = 0.47$) were similar.

Time to first capture was longer than time to subsequent captures for three sets (e.g., set 12, Fig. 2). This delay may be from (i) chance, where these timers reached bottom later than the time–depth recorder, which recorded on-bottom time, or (ii) gear encounter, bait location, and hooking sometimes taking a few minutes.

Captures most frequently were not related in both time and space, indicating that most were captured independently (Fig. 6). There were some exceptions: for example, a pair of

Table 1. Parameter estimates from the capture times.

Set	Historic model λ	Exponential model			Power model		
		λ_0	r	s	β_3	β_1	s
1	0.74	0.83	0.017	0.51	1.17	0.132	0.60
2	0.93	0.90	0.009	0.51	1.14	0.077	0.60
3	0.79	0.90	0.023	0.51	1.21	0.137	0.60
4	0.41	1.10	0.047	0.51	2.67	0.382	0.60
5	0.83	0.65	0.001	0.51	0.65	0.000	0.60
6	0.75	0.71	0.011	0.51	0.86	0.075	0.60
7	1.33	1.23	0.017	0.51	1.37	0.068	0.60
8	0.60	0.56	0.007	0.51	0.63	0.051	0.60
9	0.26	0.22	0.002	0.51	0.31	0.064	0.60
10	0.16	0.20	0.006	0.51	0.43	0.156	0.60
11	0.55	0.60	0.011	0.51	1.07	0.151	0.60
12	0.79	0.61	0.000	0.51	0.63	0.000	0.60
13	0.31	0.50	0.015	0.51	1.41	0.264	0.60
14	0.28	0.51	0.034	0.51	1.06	0.282	0.60
15	0.07	0.07	0.000	0.51	0.07	0.000	0.60
16	0.33	0.32	0.004	0.51	0.30	0.001	0.60
17	0.22	0.22	0.004	0.51	0.23	0.027	0.60
18	0.44	0.33	0.000	0.51	0.34	0.000	0.60
19	0.34	0.37	0.008	0.51	0.64	0.131	0.60
Average	0.53	0.57	0.011		0.85	0.105	
Minimum	0.07	0.07	0.000		0.07	0.000	
Maximum	1.33	1.23	0.047		2.67	0.382	
–log-likelihood	950	781			844		
Akaike information criterion	988	859			922		

Note: Sets 1–9 were located in Chatham Strait near Tenakee Inlet and sets 10–19 near Point Couverden. The hook spacing of sets 1–13 was 2 m and for sets 14–19 was 1 m. Parameter estimates are for hooks with hook timers only, which were located on every hook for 2-m-spacing skates and every other hook for 1-m-spacing skates. As a result, the estimated encounter rates for all hooks for 1-m-spacing skates is twice the listed values. λ , encounter rate; λ_0 , initial encounter rate; r , rate of decline of encounter rate; s , fraction of baits left when bait location probability began to decrease. β_3 is proportional to initial encounter rate and β_1 is a measure of feeding regimen. The Akaike information criterion measures how well the model represents the data; a lower Akaike information criterion value implies a better fit.

fish were captured nearly simultaneously at positions 37 and 38 of set 7 and may represent fish traveling together or the first capture may have cued the second capture. The spatial distribution of captures was examined to determine whether such simultaneous captures were evidence that sablefish captures were clustered. If captures were clustered, runs of adjacent captures would be longer than if they were randomly distributed. The maximum number of adjacent captures observed was six. A Monte Carlo simulation was used to test whether this maximum was unusual. The test consisted of randomly assigning capture position of the total catch, recording the maximum number of adjacent captures, and then repeating these steps 10 000 times. A maximum of six adjacent captures occurred 16% of the time, and the most frequent maximum was seven, which occurred 28% of the time. In fact, maximum runs of six or more occurred over 95% of the time. Clearly, this evidence does not support clustering of individuals along the longline. Some captures were related in time, but not in space; for example, sablefish were captured in groups at set 3 (Fig. 2), but within each group usually were separated in space (Fig. 6). Such a pattern might occur if sablefish organize spatially into successive pulses or waves of moving, feeding fish or if detection of the gear by an individual leads others to the gear and they begin searching along the line and discovering the remaining

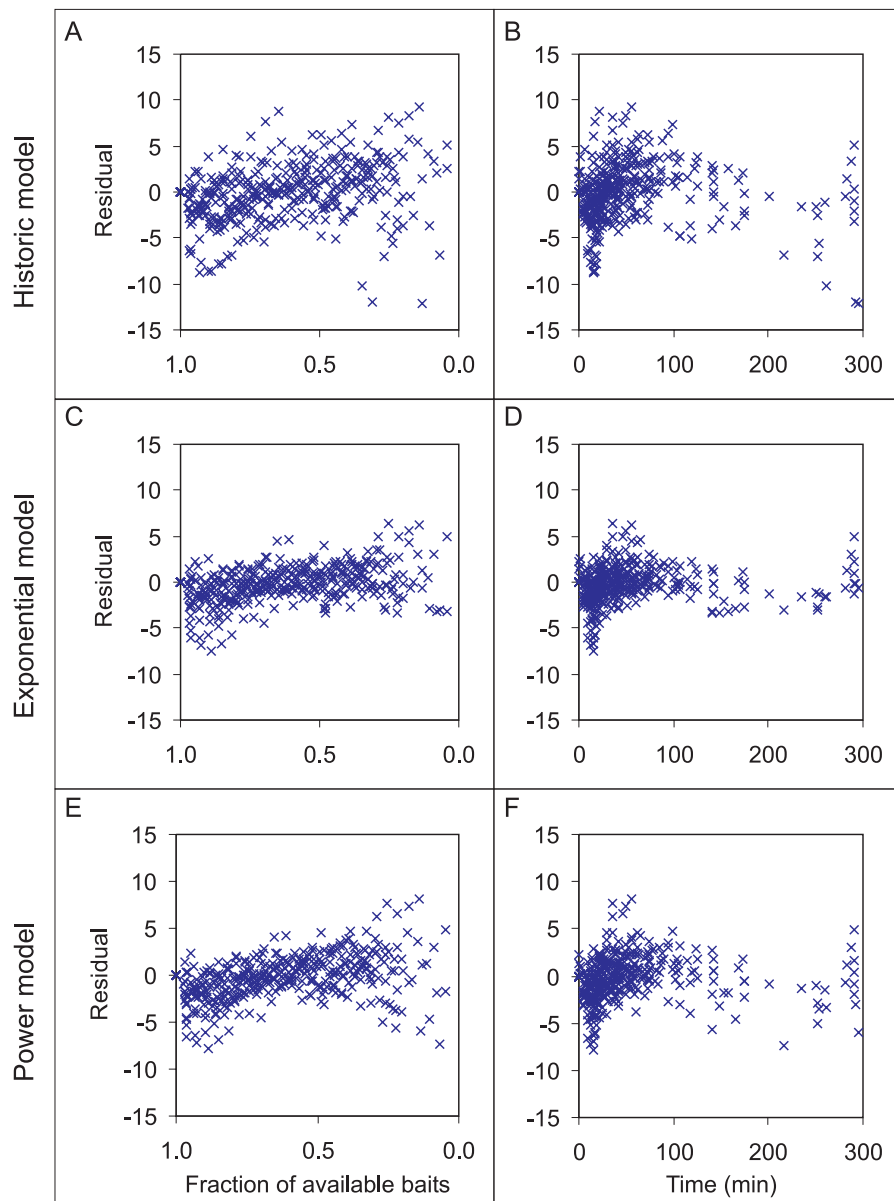
baits. Fieldwork on sablefish group feeding behavior is necessary to determine whether something other than chance caused these groupings.

Other longline gear experiments

Hook density, but not hook pattern, affected sablefish catch. Greater hook density led to greater sablefish catch per metre of line and lower catch per hook (ANOVA, $p < 0.001$) (Table 2). In contrast, catch per metre of line was similar regardless of hook pattern in the first hook pattern experiment (regression: sablefish, $p = 0.42$; spiny dogfish (*Squalus acanthias*), $p = 0.82$) (Table 2). Unlike the other experiments, spiny dogfish, not sablefish, was the primary species caught. More hook patterns were tested in the second hook pattern experiment, and catch per metre of line was found to decrease as the gap between hook clusters increased (regression: sablefish, $p = 0.01$), but the estimated decrease was small, only 8% over the tested range of 12- to 42-m gaps.

Most sablefish were caught during the first 3 h of soak time; soaking the gear an additional 4 h resulted in only slightly greater catch per hook (0.28 versus 0.24 for 3 h soak, paired t test, one-tailed, $p < 0.001$) (Table 2). Only 15% more sablefish were caught by soaking an additional 4 h (regression: 7- versus 3-h catch rates, $p < 0.001$); equivalently, the proportion caught during the first 3 h averaged

Fig. 3. Residuals (observed less expected catch) by model. (A) Residuals versus available baits for the historic model; (B) residuals versus time for the historic model; (C) residuals versus available baits for the exponential model; (D) residuals versus time for the exponential model; (E) residuals versus available baits for the power model; (F) residuals versus time for the power model. The models are described in the Mathematical model section.



0.87 of the total after 7 h. The proportion was not affected by sablefish density and was similar regardless of catch per hook after 3 h (Fig. 7). Only 3–9% of herring baits, but 28% of squid baits, remained after 7 h of soak time, probably because squid is a tougher bait. Sablefish catch per hook was about 10% greater for squid bait compared with herring bait (1988 experiment, ANOVA, $p < 0.01$).

Sablefish catch per hook for reused bait was 70% of fresh bait (0.20 versus 0.14 for reused bait, paired t test, $p = 0.01$, $n = 8$ sets). The catch rate decreased less than odor production; estimated odor production for reused squid, previously soaked 3–10 h, was no more than 34% of that for fresh bait (Fig. 8). Sablefish catch per hook was unaffected by pre-soaking (0.31 versus 0.33 for presoaked bait, paired t test, $p = 0.37$, $n = 17$ sets), probably because presoaking is diffi-

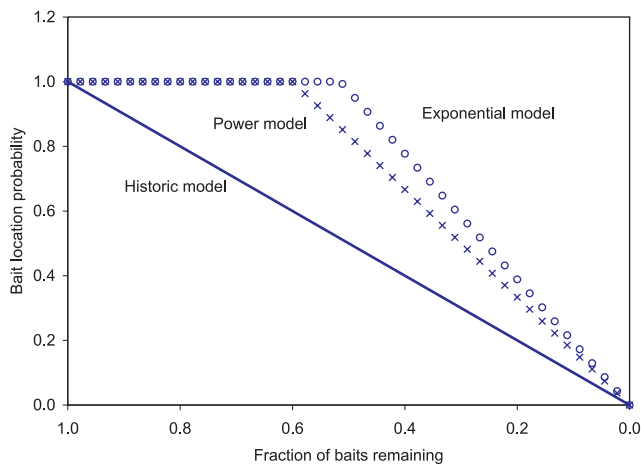
cult to replicate. Bait loses attractants sinking to 500-m water depth in about 0.5 h because sinking washes the bait of some odor and water pressure squeezes odor from the bait.

Discussion

Encounter rate

Sablefish encounter the longline gear at a decreasing rate. The hook timer estimate of encounter rate after 3 h was only 13% of the initial rate, assuming an exponential encounter rate model. In the soak time experiments, sablefish catch was only 15% higher for 7 versus 3 h. Encounter rate and bait location probability were not directly observed; only the product of encounter rate and bait location probability, the capture rate, was directly observed, so the evidence for what

Fig. 4. Bait location probability versus baits remaining for the historic (line), exponential (○), and power (×) models. The models are described in the Mathematical model section.



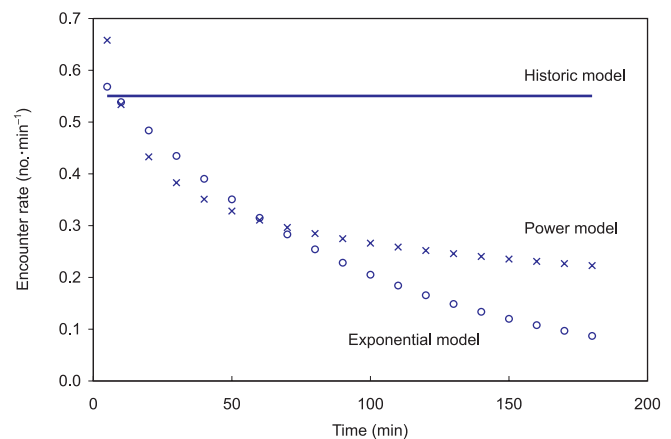
caused the soak time effect is indirect. The soak time effect was insensitive to sablefish density: the 3-h catch was a similar fraction of the 7-h catch, even though fewer baits remained in high-density areas. If the soak time effect was due to decreased bait location probability, then one would expect that the effect would be greater in areas of high sablefish density. Instead, the effect was similar, implying that decreased bait location probability did not cause the soak time effect, leaving decreased encounter rate as the most likely cause of the soak time effect.

At least two sablefish feeding behaviors, response intensity and catch rate, do not change as much as odor concentration. Sablefish response intensity decreased less than odor concentration in a bait odor experiment (Løkkeborg et al. 1995). Sablefish catch rate decreased less than estimated odor production in the reused bait experiment. The catch rate of torsk (*Brosme brosme*) also decreased less than estimated odor production in a presoaked bait experiment (Løkkeborg and Johannessen 1992).

Sablefish response to bait odor may reach a limit at high odor concentrations, which would account for the more limited range of sablefish feeding behaviors compared with odor concentration. A fish has an upper limit for turning rate, response time, swimming speed, etc., and these physical constraints may limit the range of response at higher odor concentrations. The proportion of the fish responding to the bait odor also may be limited, reaching an asymptote when the odor concentration in the plume is above a certain level, far beyond the response threshold, as occurs in the downstream area closest to the gear.

Sablefish also may balance the energy costs of feeding behavior against the likelihood of locating food once scented, limiting energy expenditure when swimming upstream to locate food. The odor concentration is low after the bait has been on bottom for a few hours; from the fish's perspective, the bait will appear as less food or as food that is distant. Sablefish, like Atlantic cod (*Gadus morhua*) and haddock (*Melanogrammus aeglefinus*), may make energy-based choices. Løkkeborg et al. (1989) found that response of both Atlantic cod and haddock to baited hooks decreased in periods of high current and hypothesized that energy optimiza-

Fig. 5. Estimated encounter rate versus time for set 11. Encounter rate was estimated for the historic (line), exponential (○), and power (×) models. The models are described in the Mathematical model section.



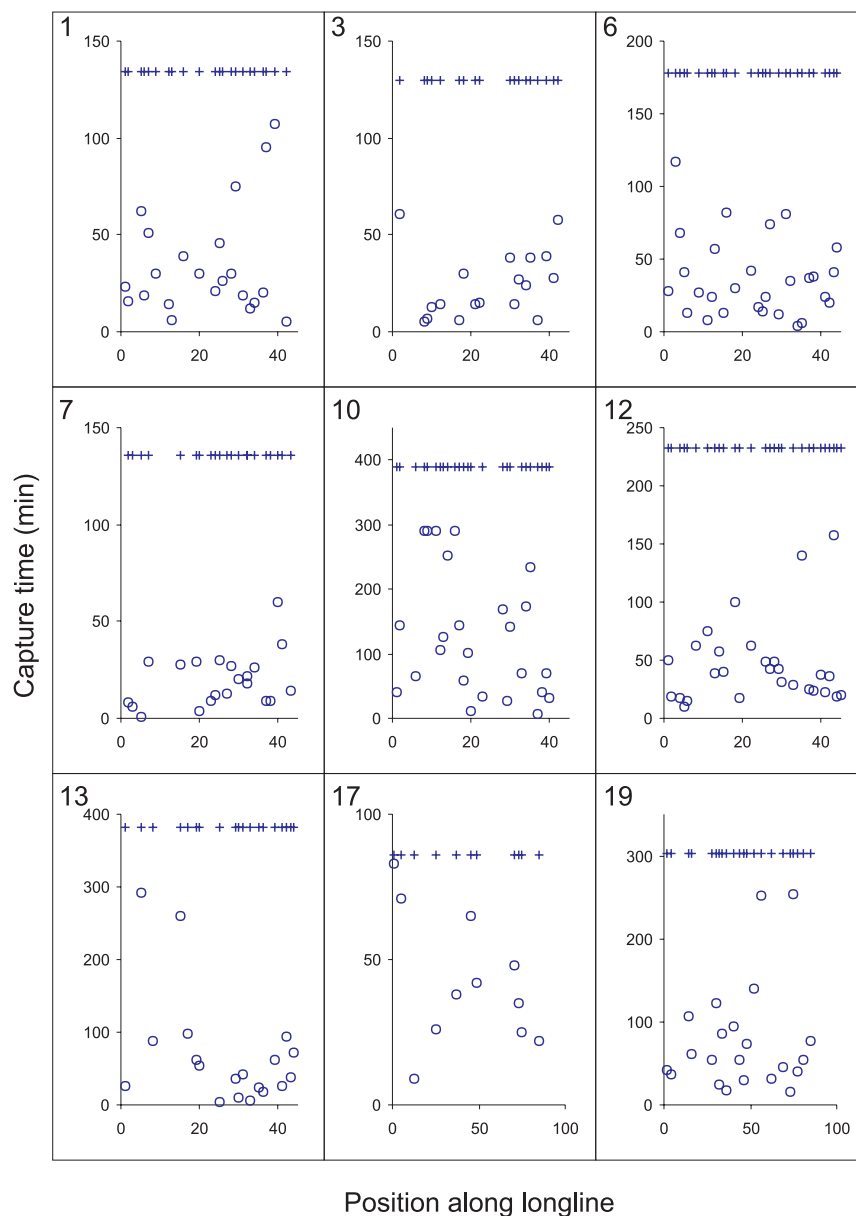
tion was achieved by restricting activity to periods with moderate or low current and staying in shelter when the current was strong.

Sablefish encounter rate appears to track odor concentration at the odor plume edge more closely than odor concentration at the bait. After 6 h, encounter rate and plume edge concentration are about 10% of the 1-h value, whereas at-bait concentration is about 40% (Fig. 9). The odor concentration at the leading edge of the plume is the odor concentration first scented by the fish. The close relationship between plume edge concentration and encounter rate indicates that this concentration is a primary determinant in the fish's decision to locate the bait, once scented.

The odor concentration behind the leading edge of the plume decreases (Fig. 10) so that fish swimming to the bait encounter decreasing odor concentration. This decrease is due to an exponential decline in odor production by the bait and to horizontal and vertical diffusion during downstream propagation of the odor plume (Sainte-Marie and Hargrave 1987). The concentration behind the plume edge may fall below the sablefish response threshold, depending on the feeding regimen. Odor concentrations for longer distances and times are below the response threshold for sablefish deprived of food for 1 day; all odor concentrations are above the response threshold for sablefish deprived of food for 4 days (Fig. 10).

Active space, the volume where the odor concentration is present in superthreshold quantities (Bossert and Wilson 1963), may shrink with time. For odor released at 1 h, the maximum length of the active space is 925 m, for 2 h is 793 m, and for 6 h is 654 m ($0.08 \text{ m} \cdot \text{s}^{-1}$ current, sablefish deprived of food for 1 day (Løkkeborg et al. 1995, their table 3)). The active space may shrink, but the scent level remains above the detection threshold of sablefish over a relatively large volume; the maximum length of the active space for odor released at 6 h was 654 m. Therefore, sufficient scent remains to attract sablefish from a wide area, but eventually, only fish whose movements bring them into contact with the active space will be attracted to the gear. Whether the active space shrinks depends on feeding regimen. The active space stops expanding after 4–5 h for sablefish de-

Fig. 6. Observed (○) capture times and cumulative captures at off-bottom time (+) versus position along longline. Only a subset of the hook timer data is displayed for brevity.



prived of food for 1 day and continues expanding for sablefish deprived of food for 4 days (Fig. 10).

Bait location probability

Sablefish are adept at locating available baits. In the hook timer experiments, the capture rate decreased slowly until as few as 20–30% of the baits remained, and the estimated bait location probability did not begin to decrease until only 50% of the baits remained. In the hook pattern experiments, the decrease in catch rate due to widely spaced and narrowly spaced baits was small, only about 8% from 12- to 42-m gaps. A 42-m gap between single baits is equivalent to only about 5% of the baits remaining for the standard survey skate with 2-m spacing. Thus, sablefish can locate widely spaced baits. This bait location ability implies that gear satu-

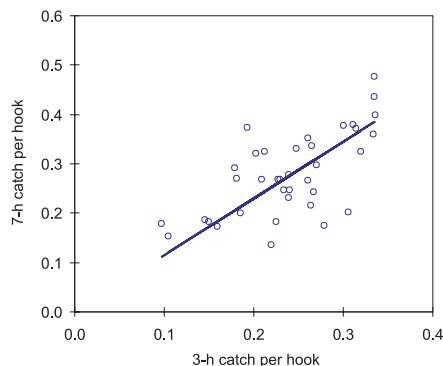
ration has little effect on the capture of sablefish, potentially until few baits remain.

Sablefish are good bait predators, probably because they are acutely sensitive to odor. A sablefish deprived of food for 1 day can detect bait odor 300–1200 m downstream of a bait soaked 1 h, depending on current; the response threshold for sablefish deprived of food for 4 days is among the lowest reported for fish (Løkkeborg et al. 1995). Adult sablefish may be this sensitive because they inhabit deep water, typically 150–1000 m, where locating prey depends on smell rather than sight. Because they are good bait predators, sablefish may behave like Atlantic cod, which Løkkeborg and Fernö (1999) observed to swim upstream directly to a baited line and then swim along the line for 100 m or more after locating the baited line. A fish with this searching be-

Table 2. Experiment, treatment, sablefish catch (number) per metre of line and per hook, baits retrieved per hook, and spiny dogfish catch per hook for seven experiments.

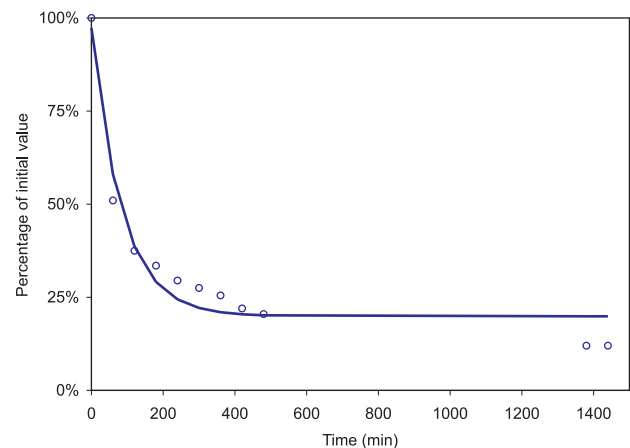
Experiment	Treatment	Sablefish catch per metre of line	Sablefish catch per hook	Baits retrieved per hook	Spiny dogfish catch per hook
Hook density, 1986 ($n = 10$ sets)	0.25 hook·m ⁻¹	0.114	0.478	0.084	0.000
	0.5 hook·m ⁻¹	0.176	0.367	0.230	0.000
	1.0 hook·m ⁻¹	0.246	0.257	0.418	0.000
Hook pattern, 1990 ($n = 18$ sets)	4-m gap	0.040	0.158	0.032	0.259
	8-m gap	0.041	0.164	0.016	0.233
	24-m gap	0.043	0.172	0.047	0.255
Hook pattern, 1998 ($n = 15$ sets)	12-m gap	0.095	0.379	0.142	0.000
	22-m gap	0.092	0.367	0.165	0.000
	32-m gap	0.091	0.365	0.164	0.000
	42-m gap	0.087	0.346	0.166	0.000
Soak time, May 1986 ($n = 6$ sets)	3 h	0.188	0.377	0.169	0.000
	7 h	0.218	0.436	0.039	0.000
Soak time, June 1986 ($n = 12$ sets)	3 h	0.140	0.281	0.213	0.000
	7 h	0.163	0.326	0.036	0.000
Soak time, 1987 ($n = 24$ sets)	3 h	0.093	0.186	0.319	0.000
	7 h	0.097	0.194	0.092	0.000
Soak time, 1988 ($n = 32$ sets)	3 h, herring	0.114	0.228	0.247	0.000
	7 h, herring	0.143	0.285	0.072	0.000
	3 h, squid	0.129	0.258	0.473	0.000
	7 h, squid	0.154	0.307	0.284	0.000

Note: The treatment labels are self-explanatory except for the hook pattern experiments. The hook patterns are all hooks spaced 4 m apart, three hooks spaced 2 m apart, 8 m of bare line, etc., five hooks spaced 2 m apart, 12 m of bare line, etc., 10 hooks spaced 2 m apart, 22 m of bare line, etc., 11 hooks spaced 2 m apart, 24 m of bare line, etc., 15 hooks spaced 2 m apart, 32 m of bare line, etc., and 20 hooks spaced 2 m apart, 42 m of bare line, etc. The hook patterns are denoted by the length of bare line between hook clusters, e.g., 12-m gap denotes five hooks spaced 2 m apart, 12 m of bare line, etc.

Fig. 7. Observed (○) and expected (line) 7-h catch rate versus 3-h catch rate for the soak time experiments.

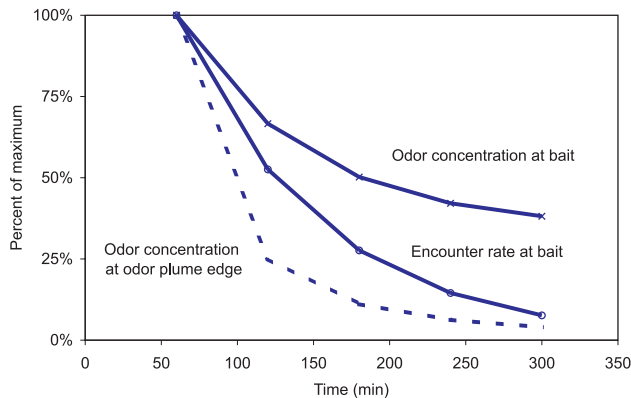
havior would probably locate an available bait unless large sections of line were occupied. Searching the line also makes biological sense as a natural food-seeking behavior. This searching behavior also implies that encounter rate is more important than locating bait.

Models relating longline catch rate to fish abundance have been developed that correct longline catch rates for gear saturation (Murphy 1960; Rothschild 1967). These early models, as well as a more recent one (Somerton and Kikkawa 1995), share two assumptions: (i) encounters to the gear are random, independent, and occur at a constant rate through time and (ii) the probability of catching fish, the bait location probability, decreases linearly with the number of baits remaining. For example, Murphy (1960) expressed the rate

Fig. 8. Observed (○) (Løkkeborg 1990, his table 1) and expected (line) rate of release of amino acids from mackerel bait, expressed as a percentage of the initial value measured at 0 h of exposure to running seawater. The equation that I used to estimate expected release rate is shown in the Appendix.

of change of the number of hooked fish as a linear function of the number of baits remaining. I estimated the bait location probability for sablefish, rather than assuming that this probability follows a particular form. The improved fit implies that for sablefish, bait location probability is not proportional to the number of baits available; instead, bait location probability remains constant, at least initially, for a wide range of available baits, probably because sablefish are

Fig. 9. Odor concentration (Sainte-Marie and Hargrave 1987; Løkkeborg et al. 1995) at bait (×) and at the odor plume edge (broken line) and encounter rate at bait (○) expressed as a percentage of values after 1 h on bottom. The calculation equation for odor concentration is shown in the Appendix and was computed assuming a current of $0.08 \text{ m} \cdot \text{s}^{-1}$ and the estimated decline in odor release rates (Fig. 8). The encounter rate was computed using the average r estimated from the hook timer experiments (Table 1) and the exponential model.

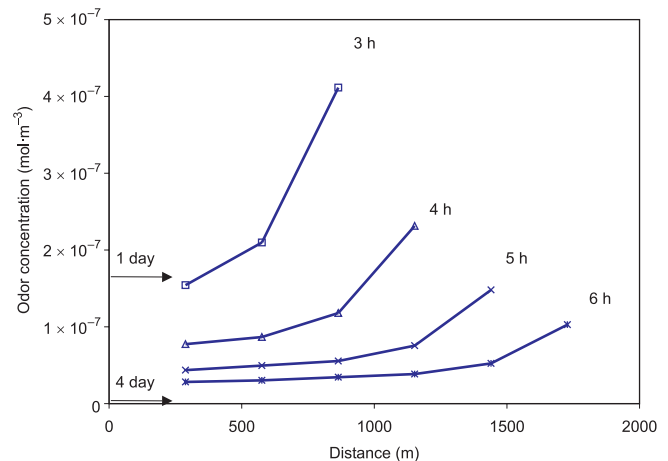


good bait predators. Different species were modeled previously: yellowfin tuna (*Thunnus albacares*) by Murphy (1960) and Rothschild (1967) and pelagic armorhead (*Pseudopentaceros wheeleri*) by Somerton and Kikkawa (1995). The bait location probability may differ by species and gear configuration, but I suspect that most species readily locate baited hooks, even when few remain. If so, gear saturation of longlines may be less important than previously considered.

Hooking fraction

Sablefish encountering the gear may interact, but any interaction appears inconsequential. The random distribution of sablefish captures along the line implies that initial captures do not affect subsequent captures, so that hooking fraction appears unaffected by interaction between individuals. The information supporting this conclusion is indirect because hooking fraction, like encounter rate and bait location probability, was not directly observed. For other species, interaction between individuals has been shown to both enhance and reduce captures. Captures of Pacific halibut (*Hippoglossus stenolepis*) tended to be clustered based on the observed frequency of adjacent captures (Skud and Hamley 1978). Cuing on captured fish or a contagious distribution of fish while free-swimming may cause clustering. Fernö et al. (1986) observed that aggressive interactions occurred when several whiting were in the vicinity of the line. Løkkeborg and Bjordal (1992) observed large Atlantic cod frightening smaller Atlantic cod away from baited hooks and torsk chasing smaller fish away from baited hooks. Interspecific interactions can also increase the hooking fraction. Responses by Atlantic cod were often initiated by a struggling haddock that had just been hooked (Løkkeborg et al. 1989). In addition, behavioral interactions may increase feeding success. Foraging success of ephemeral food patches is enhanced in groups of juvenile walleye pollock (*Theragra chalcogramma*) compared with isolated fish (Baird et al.

Fig. 10. Concentration profiles of bait odor by distance from the bait and on-bottom time (Sainte-Marie and Hargrave 1987; Løkkeborg et al. 1995) assuming a current of $0.08 \text{ m} \cdot \text{s}^{-1}$ and the estimated decline in odor release rates (Fig. 8). The calculation equation for odor concentration is shown in the Appendix. For example, after 3 h on bottom, the leading edge of the plume is about 800 m from the bait and the odor concentration there is about $0.000\,000\,4 \text{ mol} \cdot \text{m}^{-3}$, whereas the odor concentration 500 m from the bait is less, e.g., about $0.000\,000\,2 \text{ mol} \cdot \text{m}^{-3}$. The arrows indicate the response threshold for sablefish deprived of food for 1 day and sablefish deprived of food for 4 days (Løkkeborg et al. 1995, their table 3).



1991). Schooling chum salmon (*Oncorhynchus keta*) juveniles gain information by interpreting conspecific behavior (Ryer and Olla 1991).

Interaction between individual sablefish appears inconsequential, but other factors such as season, fish density, and food abundance may affect sablefish hooking fraction and thus the proportion of fish caught. Observed hooking fractions for Atlantic cod and haddock were higher in September than in December, possibly due to seasonal variation in feeding activity; feeding activity is reduced during the spawning period, which includes December (Løkkeborg et al. 1989). Fish density also may affect hooking fraction. The initiation of Atlantic cod responses by struggling haddock (Løkkeborg et al. 1989) implies that captures may serve to motivate fish to locate and attack baits even when few baits remain. Greater numbers of responding and hooked fish increase the probability of intensified responses by other fish through social facilitation and may counteract reduced bait location probability due to fewer available baits. Food abundance may vary annually or be reduced by high fish density and also may affect hooking fraction. Reduced food abundance may lead to increased feeding motivation (sablefish response to bait odor was influenced by feeding frequency (Løkkeborg et al. 1995)), intensified responses towards the baits, and increased hooking fraction.

Hooking fraction may differ between sizes of sablefish. For example, large Atlantic cod frightened smaller Atlantic cod away from baited hooks, torsk chased smaller fish away from baited hooks (Løkkeborg and Bjordal 1992), and larger fish may be attracted to the baits from farther away. As a result, longline gear may not provide accurate estimates of all

size-classes due to size-based differences in the proportion caught. However, the probability that an individual fish will be caught once in contact with the sampling gear varies with size for most species and gear types (Gunderson 1993).

Hook density

Hook density and spacing were varied in the hook density experiments, whereas in the hook pattern experiments, hook spacing was varied, but overall hook density was constant. Sablefish catch per metre of line increased with hook density but remained nearly constant with hook pattern. The lack of a substantial pattern effect implies that the greater number of hooks, not their closer spacing, increased sablefish catch in the hook density experiments.

A larger active space is probably responsible for the relationship between catch and hook density. A higher hook density increases odor concentrations within the odor plume formed downstream of the gear. A denser odor plume enlarges the active space and therefore attracts more fish. A denser odor plume may also increase the hooking fraction. Scent concentration affects sablefish response intensity (Løkkeborg et al. 1995) and thus may increase the feeding motivation.

Varying hook spacings from 2.7 to 12.8 m also affected Pacific halibut catches (Skud and Hamley 1978). As with sablefish, higher hook densities resulted in significantly increased catch per unit of line and decreased catch per hook of Pacific halibut (Skud and Hamley 1978). They addressed the differences in "catch per hook" and suggested that these differences are due to competition between hooks for the available fish: if more than one hook falls within a Pacific halibut's range, these hooks "compete" for the capture of that fish, and the closer they are spaced, the lower the catch per hook is; on the other hand, if the hooks are so far apart that each Pacific halibut encounters no more than one hook, then the hooks fish independently and their efficiency is at its maximum, so that catch per hook no longer increases with hook spacing. Eggers et al. (1982) developed this concept to describe a quantitative method for estimating the effective radius of a baited hook or pot, assuming that each device has on average a capturing field of fixed size and that the capturing field is a circle centered at the baited hook or pot. They applied this method to three species, including Pacific halibut and sablefish. Both methods (Skud and Hamley 1978; Eggers et al. 1982) assume that the capture field per hook is constant. Skud and Hamley (1978, p. 20) phrased this assumption as "...assuming a concentration of bait [does not] attract halibut beyond their normal feeding ranges...." However, "catch per meter of line" increased for halibut and sablefish, implying that the capture field per hook was not constant, but rather, increased with hook density. The capture field per hook may increase because odor plumes from adjacent baits overlap, increasing the downstream odor concentrations and intensifying fish response. If so, the capture field is not constant as Skud and Hamley (1978) and Eggers et al. (1982) assumed, and their measuring the capture field of a hook by experimentally varying hook spacing is not valid.

Conceptualization of sablefish capture by longline and implications for longline survey design

Encounter rate, not bait location probability, appears to

control sablefish capture by longline. Sablefish encounter the gear at a decreasing rate with on-bottom time. The encounter rate decreases, but not their ability to locate baits. Sablefish locate baits even as the number of available baits decreases substantially. The bait location probability is constant, at least for a substantial range of available baits, and is not a linear function of the number of available baits, as assumed by the models of Murphy (1960) and Rothschild (1967).

Encounter rate decreases with time, presumably in response to decreasing odor concentration. Odor concentration is highest at the leading edge of the odor plume and decreases behind the leading edge. Odor concentration at the leading edge of the odor plume, the concentration first encountered by the fish, appears to control the encounter rate. The relationship of encounter rate and odor concentration is modulated by fish behavior. The odor concentration decreases rapidly with time, both at and behind the leading edge of the odor plume, yet response intensity and catch rate do not exhibit the same range as odor concentration. Physical constraints or energetic considerations may limit the response range of individual fish. The proportion of the fish responding to the bait odor may approach a limit asymptotically.

Hooking fraction appeared unaffected by interactions between individual sablefish but was not observed directly and may vary in other situations. Hooking fraction has been shown to be influenced by interaction between individuals, season, and conceivably fish density. Other factors such as temperature and food availability also could vary between years and influence the proportion of fish that are caught.

The value of power function parameter β_1 estimated from the capture times can be compared with the slope parameter relating sablefish response intensity and odor concentration estimated by Løkkeborg et al. (1995), who found that the slope parameter value was related to feeding regimen and was 0.397 and 0.163 for 1 and 4 days of feeding deprivation. The value of β_1 estimated from the capture times of the hook timer experiments was 0.105, implying that sablefish in the study areas had not fed for more than 4 days or an equivalent feeding regimen.

Survey catch rates likely provide an accurate linear index of relative abundance over a wide range of sablefish density because sablefish appear adept at locating available baits and because the encounter rate decreases with time. Capture rate slows and most sablefish are caught within the survey's minimum soak time of 3 h; the survey soak time varies from 3 to 10 h because of the length of time required to retrieve the gear. This slowing appears due to reduced encounter rate rather than an inability to locate the decreasing number of baits remaining. This slowing also appears insensitive to sablefish density; catches for longer soak times are similar to the 3-h catch because of the slowed encounter rate. The weak link between catch rate and abundance is the unknown extent that hooking fraction and other factors such as temperature and food availability vary and influence the proportion of fish caught.

Findings of these experiments imply that longline surveys can be designed for a linear relationship between survey catch rate and abundance over a wide range of fish density. The gear should be on bottom long enough that the encoun-

ter rate has slowed sufficiently for most fish to be caught. This minimum on-bottom time can be determined by a series of soak time experiments to learn how catch rate varies with soak time. In addition, the bait density should be high enough that some baits remain after the maximum soak time, so that an available bait can still be located. This bait density can be determined by increasing hook density if "few" baits are left at the maximum soak time to ensure that the decrease in catch rate with soak time is due to decreased encounter rate rather than too few baits remaining. A well-designed longline survey will have enough soak time to include most captures and enough baits for some to remain at the end.

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Appendix

The power encounter rate model combines three functions: sablefish response intensity as a function of odor concentration, the odor concentration at the leading edge of the odor plume, and the length of time to travel from the leading edge to the bait. Løkkeborg et al. (1995) measured sablefish response intensity, RI (number and frequency of turns), to different odor concentrations, C (molar), in a laboratory and found that $\log_{10} RI = \beta_0 + \beta_1 \log_{10} C$; the values of the regression coefficients depended on the feeding regimen.

Odor concentration, derived from the active-space equation in Sainte-Marie and Hargrave (1987) and Løkkeborg et al. (1995), is

$$C = \frac{QU}{x^2 2\pi B_y B_z}$$

where Q is the release rate of the odor source (moles per second), U is the current (metres per second), x is downstream distance (metres), and B_y and B_z are the horizontal and vertical dispersion velocities (metres per second). I used the same values of the dispersion velocities, i.e., $B_y = 10^{-3} \text{ m}\cdot\text{s}^{-1}$ and $B_z = 10^{-5} \text{ m}\cdot\text{s}^{-1}$, as Sainte-Marie and Hargrave (1987) and Løkkeborg et al. (1995) and similarly assumed that the relationship between horizontal and vertical standard deviations and downstream distance was linear. An equation for the release rate of the odor source is shown in the last paragraph of the Appendix.

The downstream propagation of the odor plume depends on advection (Sainte-Marie and Hargrave 1987), and its length at time t is the product of current v and time, i.e., $x(t) = vt$. However, fish arriving at bait at t are probably at-

tracted from much shorter distances because they also must swim upstream to the bait after contacting the plume. The actual distance of attraction at t is $x_A(t) = U(S - U)t/S$ (Sainte-Marie and Hargrave 1987), where x_A is the attraction distance and S is fish swimming speed (metres per second). The swimming speed must be greater than current if the fish is to reach the gear (the equation applies only if $S > U$).

I combined these three equations to form an expression for the response intensity of fish arriving at the bait at time t :

$$RI(t) = 10^{\left(\beta_0 + \beta_1 \log_{10} \frac{QU}{(U(S-U))^2 2\pi B_z B_z} \right)} 10^{\beta_1 \log \frac{1}{t^2}}$$

Fish swimming speed and feeding regimen and current are unknown and cannot be estimated separately from the experimental data, so I combined the first part of the equation, the part that is not time dependent, into a single parameter:

$$RI(t) = \beta_2 10^{\beta_1 \log \frac{1}{t^2}}$$

Assuming that encounter rate is proportional to response intensity:

$$\lambda(t) = \lambda_0 \beta_2 10^{\beta_1 \log \frac{1}{t^2}}$$

where λ_0 is the initial encounter rate. The initial encounter rate and β_2 cannot be estimated separately from the experimental data, so a single parameter $\beta_3 = \lambda_0 \beta_2$ is estimated. The parameter β_3 is proportional to the initial encounter rate and t represents capture time for the hook timer experiments. The parameter β_1 is one of the coefficients of the response intensity regression estimated by Løkkeborg et al. (1995),

who found that its value was a function of sablefish feeding regimen.

An exponential model was used to represent the release rate of the odor source:

$$R_u = R_0 e^{\psi u} + \Omega$$

where R_u is the release rate of amino acids at time u (minutes), ψ is a parameter that describes the decline of the release rate, and Ω is a parameter that describes the y asymptote. Observations of the rate of release of amino acids from mackerel bait (Løkkeborg 1990, his table 1), expressed as a percentage of the initial value measured at 0 h of exposure to running seawater, were used to estimate model parameters using nonlinear least squares and assuming additive error. The observed rate was averaged from the two analytical techniques used. The rate of decline of amino acid release for mackerel bait $\psi = 0.012$. The amino acid release rate reached an asymptote ($\Omega = 19.9\%$ of the initial release rate), indicating that the bait continues to release amino acids for a long time, at least 24 h. The lengthy release period may be due to the thickness of the bait, with odor continuing to diffuse from the interior of the bait to be released.

Appendix references

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